

Docker and the end of the distro

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Cloud native networking

TL;DR

A great part of the Docker success story has been reuse of existing distros

This is a bad thing, and we can do better

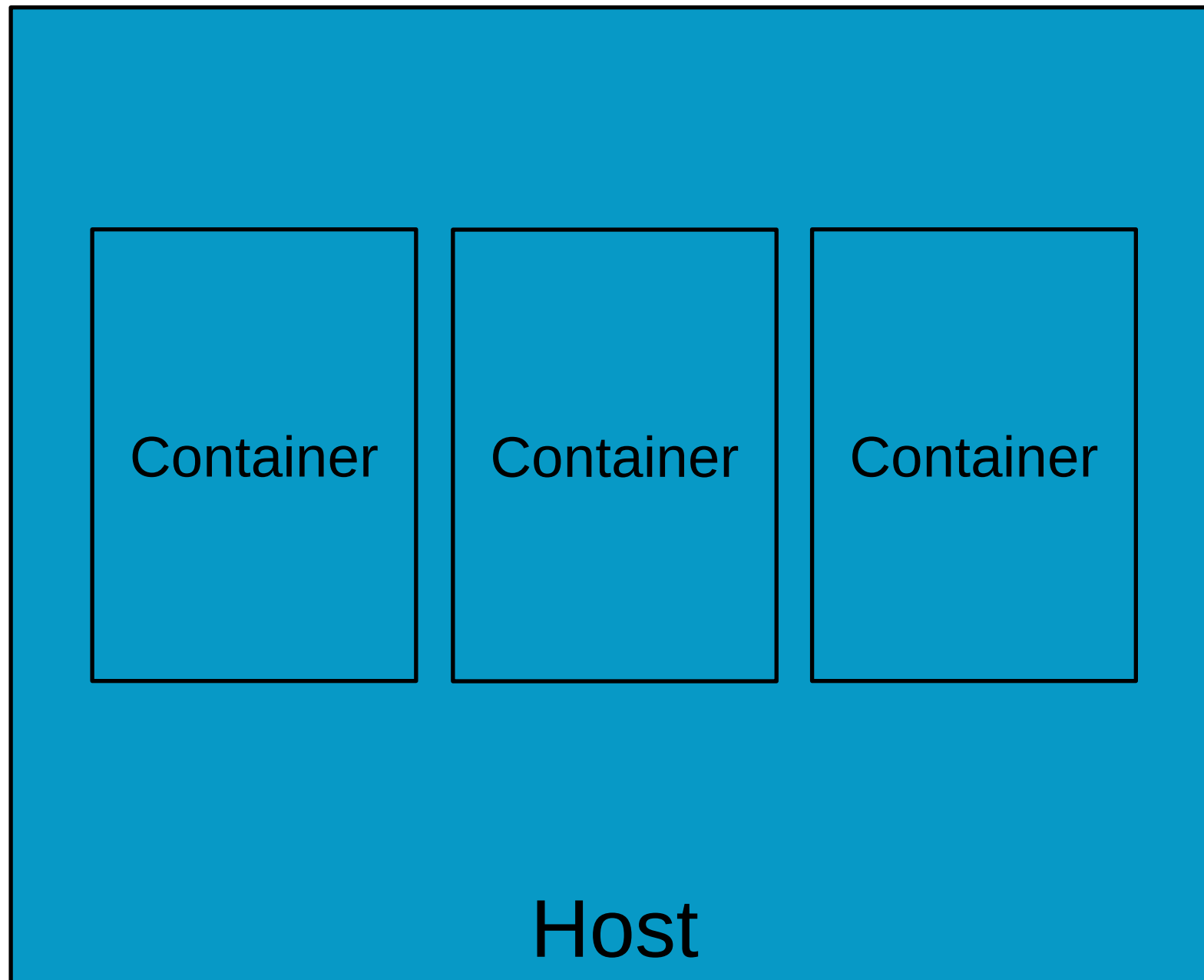
We've already done something for what we run Docker on.

Next up – what we run in Docker

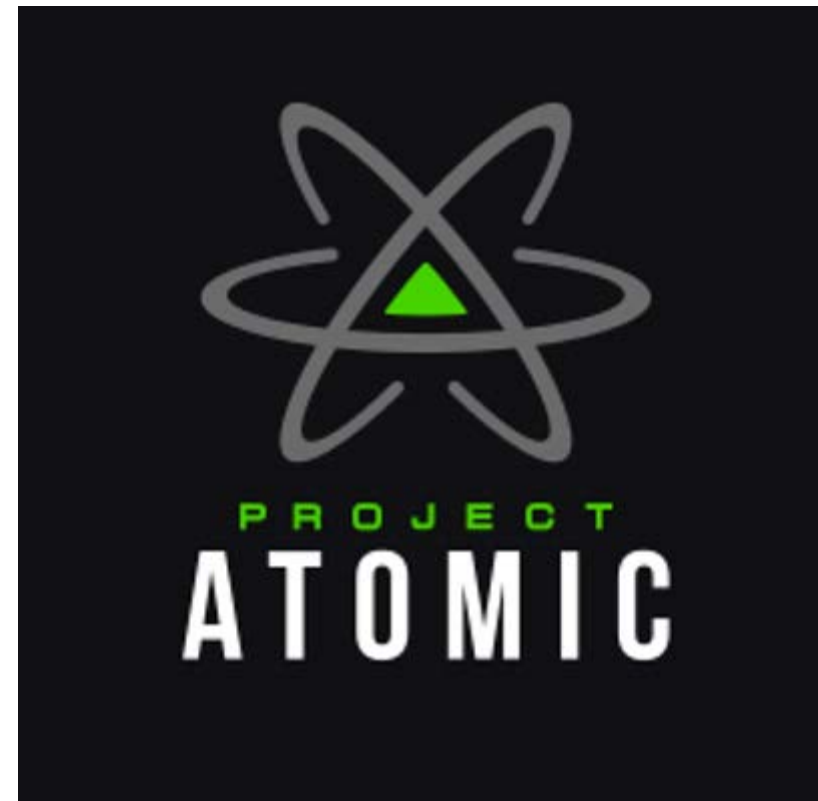
Docker runs just fine in/on



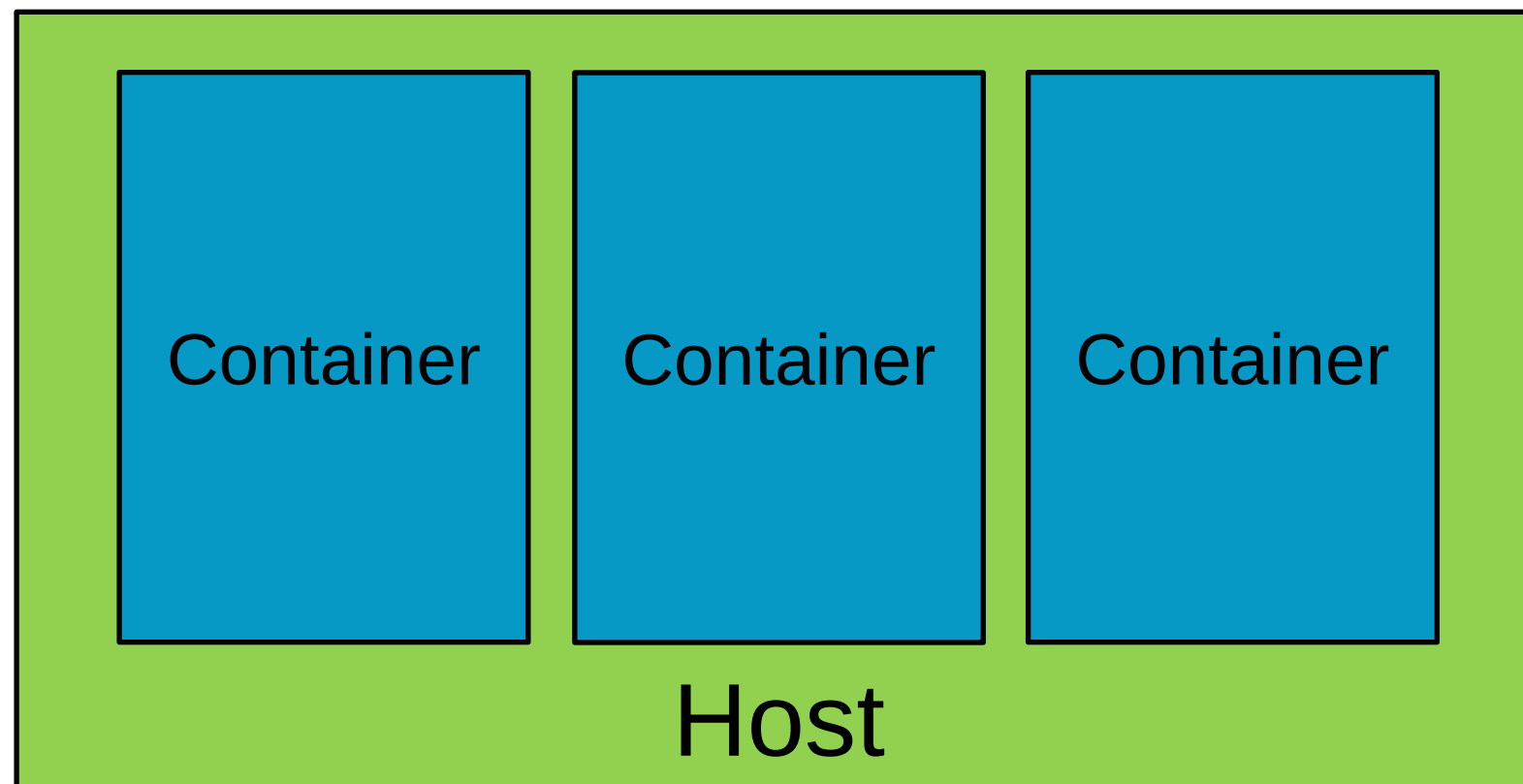
So quite often
host OS == container OS



We don't need a full OS to run Docker



So now we have
host OS < container OS



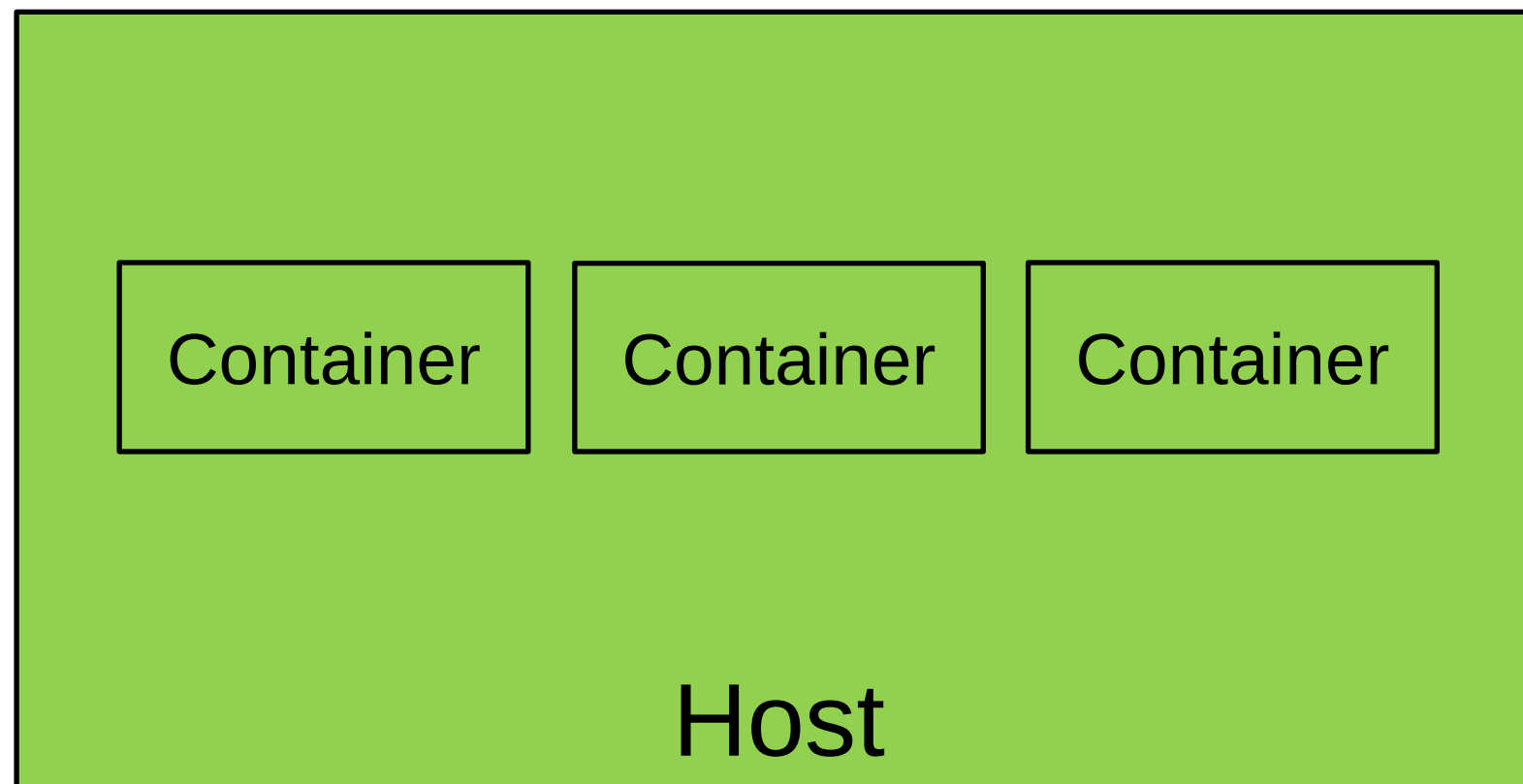
What we run Docker on != what we run in Docker



CoreOS



So we can't easily get this



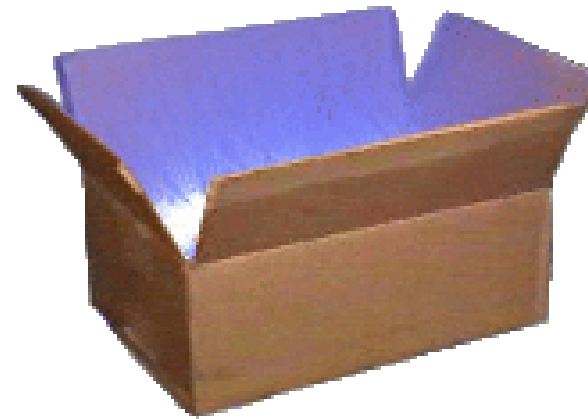
But we don't need a full OS inside containers

 **debian**

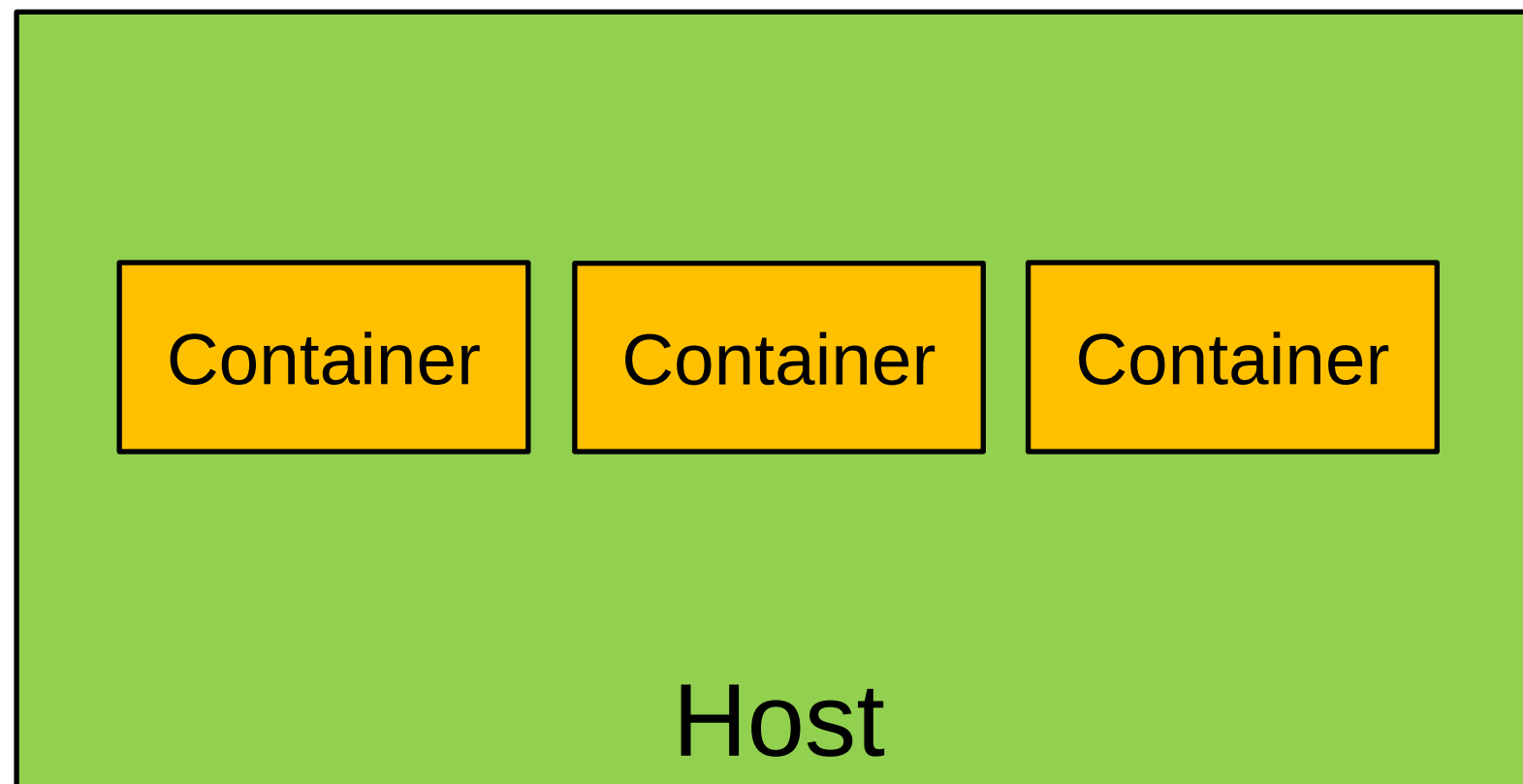
 **centos**



ubuntu



But we might aspire to this



The manifest problem

A typical Dockerfile

branch: master

[dockerfiles](#) / [meetup](#) / **Dockerfile**



Chris Swan 2 minutes ago Dockerfile example for Chicago Docker Meetup

0 contributors

file 18 lines (17 sloc) 0.513 kb

Open

Edit

Raw

Blame

History

Delete

```
1 FROM ubuntu:12.04
2
3 MAINTAINER cpswan
4 # Add universe repository to /etc/apt/sources.list
5 # we need it for nginx
6 RUN sed -i s/main/'main universe'/ /etc/apt/sources.list
7 # Update repos so that changes can take effect
8 RUN apt-get update
9 # Install nginx
10 RUN apt-get install -y nginx
11 # Turn off daemon mode
12 RUN echo "\ndaemon off;" >> /etc/nginx/nginx.conf
13 # Meetup customisation
14 RUN sed -i s/nginx/'Chicago Docker Meetup'/ /usr/share/nginx/www/index.html
15 # Expose web server
16 EXPOSE 80
17 # Run nginx
18 CMD /usr/sbin/nginx
```

Each active line creates a layer

branch: master

[dockerfiles](#) / [meetup](#) / **Dockerfile**



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```

Base OS

Sources

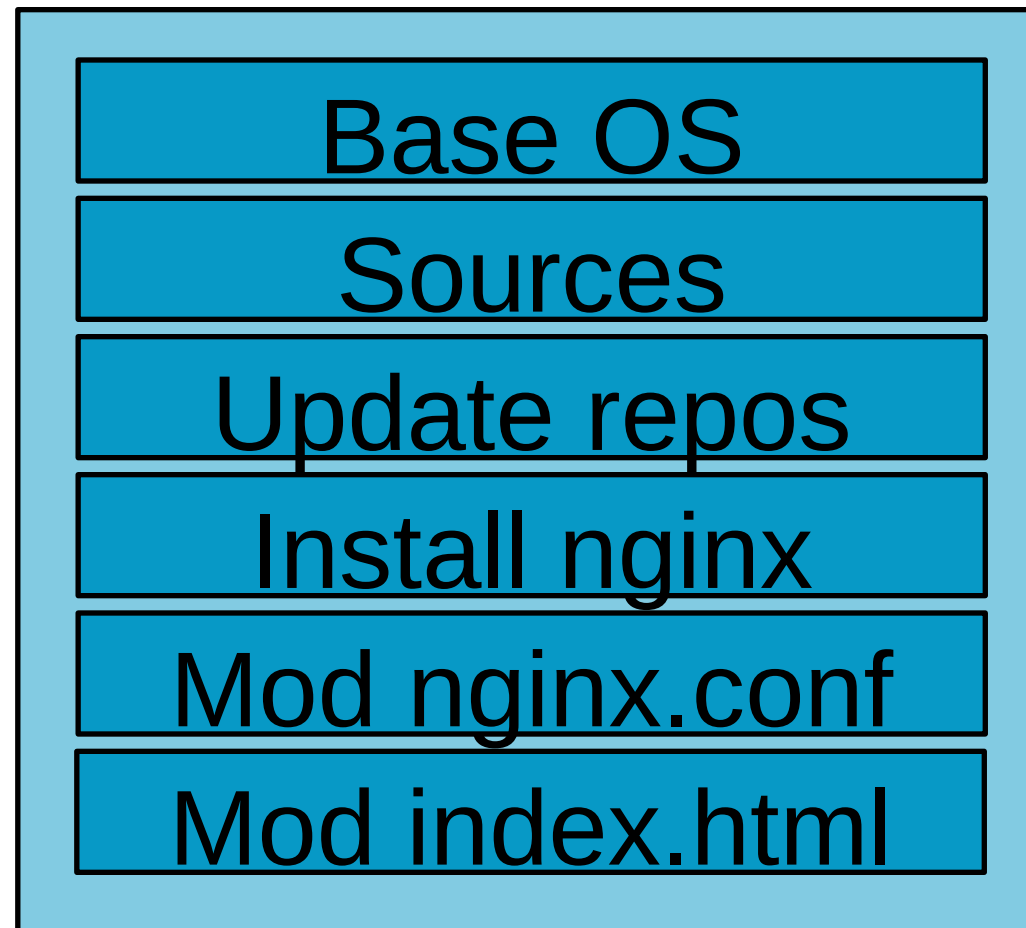
Update repos

Install nginx

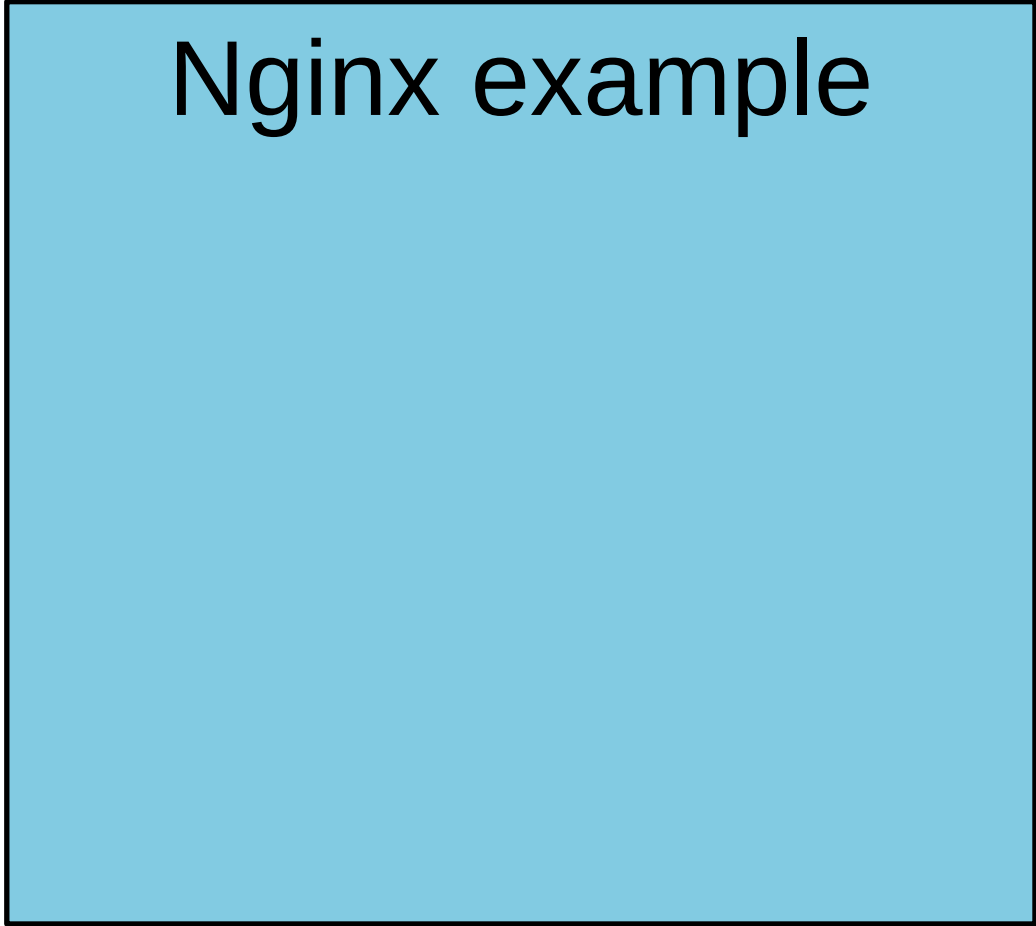
Mod nginx.conf

Mod index.html

An image binds layers together

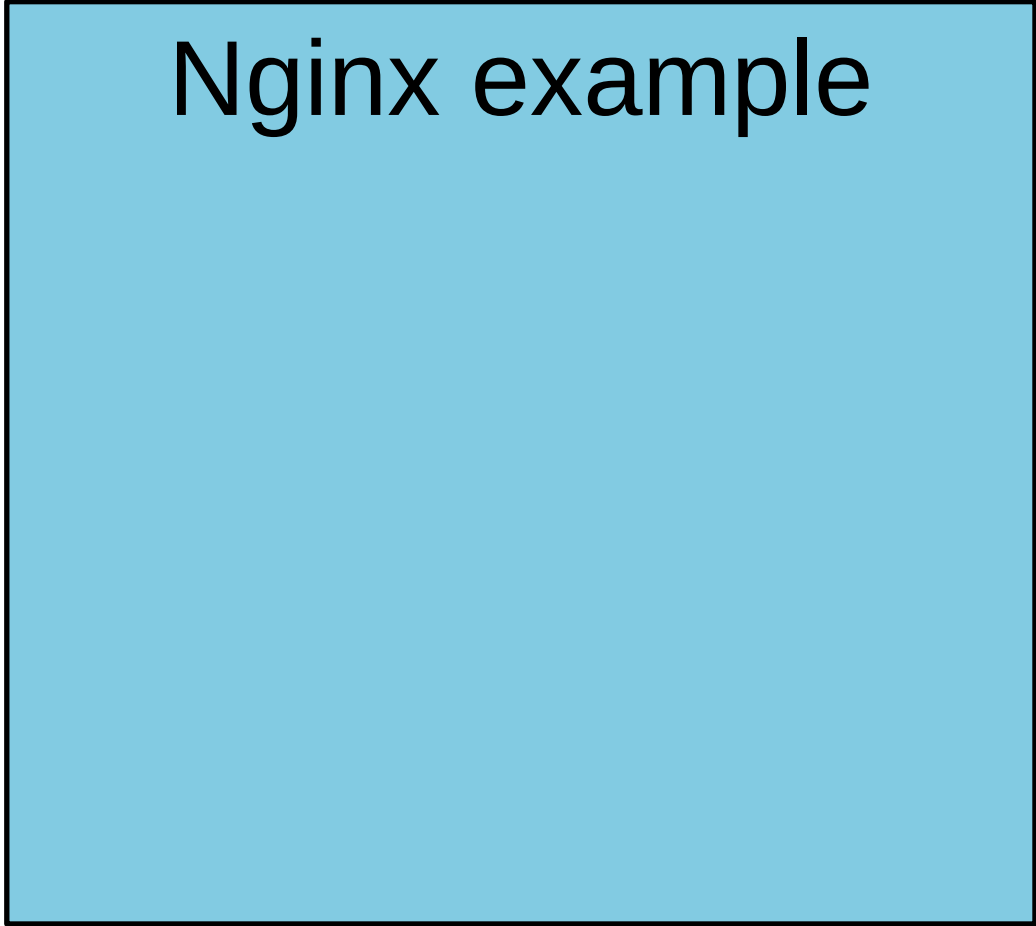


The image is the unit of deployment



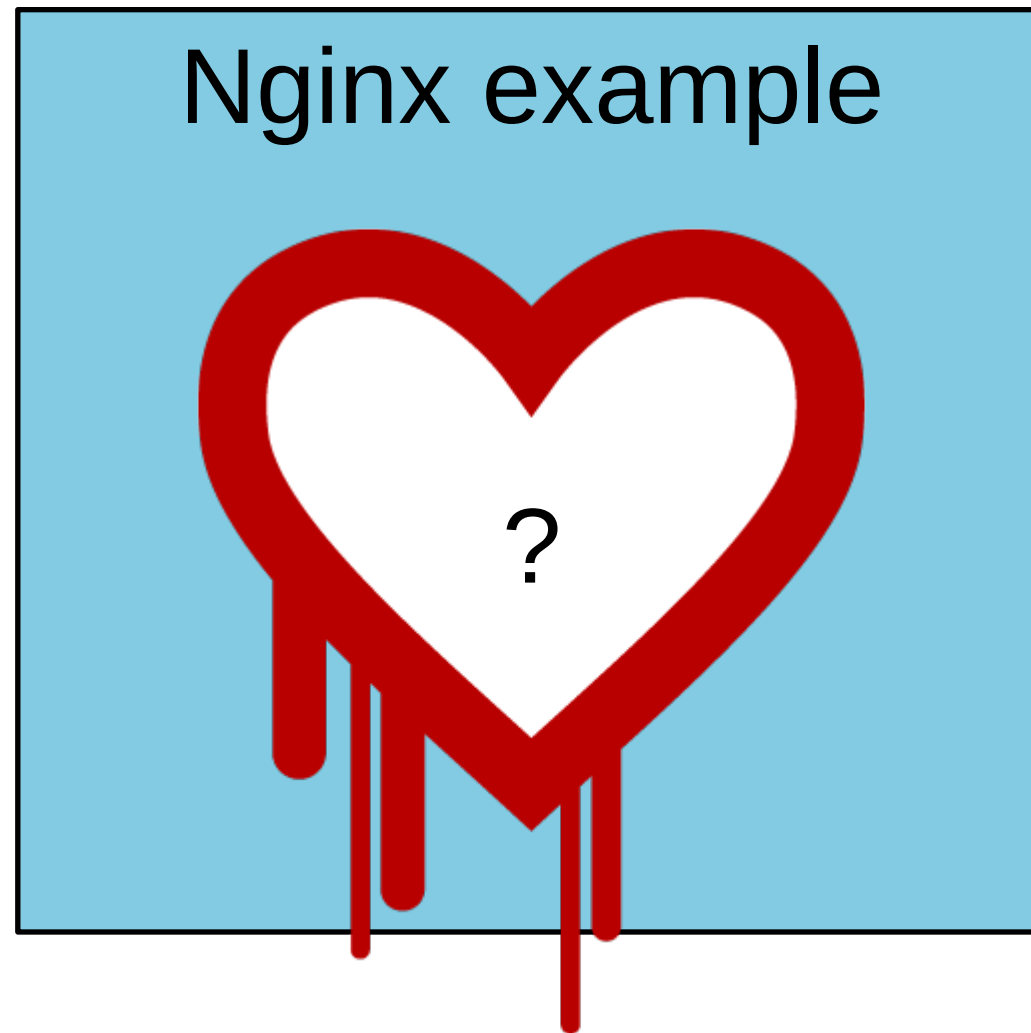
Nginx example

What version of nginx is that?



Nginx example

What version of OpenSSL installed?



and which bash?

Nginx example



Problem 1 – non determinism

Whilst we want this to be cached in the short term:

```
apt-get install nginx
```

We perhaps don't want it cached in the long term

What are those durations?

Dockerfile Jenga



CC Image credit: Ed Garcia <https://www.flickr.com/photos/egarc2/2432270195>

2 – the manifest problem

When I run

```
apt-get install nginx
```

I don't know which version of nginx I just got

Should I?

```
nginx -v > some_log.txt
```

Or maybe?

```
apt-cache policy nginx > some_log.txt
```

Recommendation 2 (of 6)

Get specific:

```
1 RUN apt-key adv --keyserver pgp.mit.edu --recv-keys 573BFD6B3D8FBC641079A6ABABF5BD827BD9BF62
2 RUN echo "deb http://nginx.org/packages/mainline/debian/ wheezy nginx" >> /etc/apt/sources.list
3
4 ENV NGINX_VERSION 1.7.7-1~wheezy
5
6 RUN apt-get update && apt-get install -y nginx=${NGINX_VERSION}
```

But...

might still pull in a bunch of point in time dependencies

Check out <http://container-solutions.com/2014/11/6-dockerfile-tips-official-images/> (<http://is.gd/sixtips>)

Solomon promises to fix things



Solomon Hykes
@solomonstre



Following

@cpswan @nbartlett @rsdunne big changes coming. Self-describing images, content-addressable layers, end-to-end signature, hermetic builds.

FROM scratch

7 lines (6 sloc) | 0.165 kb

Raw

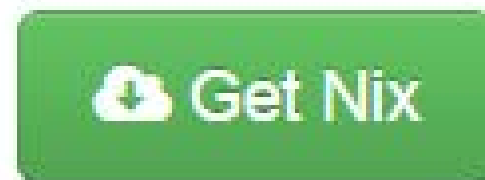
Blame

History



```
1 FROM scratch
2 MAINTAINER Zettio weave@zett.io
3 WORKDIR /home/weave
4 ADD ./weaver /home/weave/
5 CMD []
6 ENTRYPOINT ["/home/weave/weaver", "-iface", "ethwe", "-wait", "5"]
```

There is another way



TL;DR

A great part of the Docker success story has been reuse of existing distros

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We've already done something for what we run Docker on.

Next up – what we run in Docker

Questions?

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